

SYTOMINO, Russian Federation

WMO: 238470

Lat: 61.3019N

Lon: 71.3022E

Elev: 33

StdP: 100.92

Time Zone: 5.00 (E05)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.8	-38.0	-43.6	0.1	-40.3	-40.8	0.1	-37.7	8.3	-15.9	7.5	-14.4	1.5	300	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	28.7	19.1	26.9	18.5	25.1	17.7	20.7	26.7	19.7	25.1	18.6	23.4	2.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	23.5	17.6	12.6	22.6	16.6	11.9	21.9	59.8	26.7	56.5	25.0	53.1	23.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.3	6.4	5.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-43.9	21.8	1.9	0.9	-45.3	22.4	-46.4	22.9	-47.5	23.4	-48.8	24.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-21.0	-16.9	-9.2	-1.5	6.7	16.1	18.8	14.6	8.2	-0.3	-13.3	-17.8
	DBStd	15.15	9.24	9.22	7.32	5.42	5.70	4.69	3.82	3.84	3.85	5.07	8.21	9.38
	HDD10.0	4763	960	752	597	345	137	7	1	7	79	319	699	862
	HDD18.3	7229	1218	985	855	594	363	97	41	126	304	577	949	1120
	CDD10.0	674	0	0	0	0	36	190	273	150	24	1	0	0
	CDD18.3	99	0	0	0	0	3	30	56	11	0	0	0	0
	CDH23.3	792	0	0	0	0	29	239	455	69	0	0	0	0
	CDH26.7	172	0	0	0	0	3	49	112	7	0	0	0	0
Wind	WSAvg	2.7	2.6	2.5	2.8	3.0	3.0	2.8	2.5	2.4	2.6	2.9	2.7	2.6
Precipitation	PrecAvg	784	50	41	55	54	58	79	75	88	69	78	76	59
	PrecMax	963	88	78	133	86	124	124	203	153	128	132	135	98
	PrecMin	617	18	8	5	5	19	46	30	40	27	31	25	25
	PrecStd	82	21	16	25	18	23	22	34	25	24	26	24	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	1.4	4.3	12.7	26.9	30.5	31.9	27.6	20.5	13.8	1.6	-0.3
		MCWB	-0.6	0.2	0.9	7.7	16.5	20.4	21.2	20.1	14.7	9.3	0.6	-1.1
	2%	DB	-3.0	-0.1	2.7	9.3	22.9	28.1	29.7	25.2	17.7	11.1	0.4	-2.1
		MCWB	-3.7	-0.8	0.1	4.9	14.1	18.7	20.1	18.5	13.0	8.2	-0.3	-2.6
	5%	DB	-6.2	-2.7	1.4	7.2	19.6	26.2	27.7	23.4	15.6	8.7	-0.6	-3.8
		MCWB	-6.7	-3.5	-0.5	3.8	12.8	17.9	19.0	17.6	12.0	6.6	-1.2	-4.4
	10%	DB	-8.5	-5.0	0.2	5.4	16.2	24.1	26.1	21.5	13.9	6.4	-2.5	-6.3
		MCWB	-8.9	-5.7	-1.4	2.7	10.4	17.0	18.1	16.5	11.1	4.7	-3.2	-6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	0.6	1.3	9.1	17.0	21.1	22.4	21.2	15.4	9.9	1.0	-0.8
		MCDB	0.2	1.2	3.2	13.3	24.6	27.8	30.0	26.2	19.5	13.0	1.5	-0.3
	2%	WB	-3.6	-0.9	0.6	5.7	15.0	20.1	21.1	19.6	13.8	8.3	0.0	-2.6
		MCDB	-2.9	-0.1	2.2	7.5	22.0	26.0	27.6	23.8	16.9	10.9	0.4	-2.2
	5%	WB	-6.6	-3.3	-0.3	4.1	12.8	19.1	20.1	18.1	12.6	6.7	-1.2	-4.3
		MCDB	-6.2	-2.6	1.4	7.1	18.8	23.9	26.0	22.0	15.0	8.4	-0.6	-3.7
	10%	WB	-8.9	-5.6	-1.3	2.9	11.0	17.9	19.2	17.1	11.4	4.9	-3.2	-6.7
		MCDB	-8.4	-5.0	0.3	5.3	15.6	23.0	24.3	21.2	13.6	6.1	-2.5	-6.3
Mean Daily Temperature Range		MDBR	7.8	8.5	9.8	8.5	9.2	9.0	9.1	8.7	6.6	4.9	7.5	7.5
	5% DB	MCDBR	10.0	9.1	8.6	10.5	13.4	11.7	11.8	11.5	9.5	7.5	7.7	7.2
		MCWBR	9.5	8.8	6.6	7.0	6.7	5.1	4.7	5.9	5.7	4.9	7.5	6.9
	5% WB	MCDBR	9.9	9.1	8.1	9.2	12.4	9.8	10.9	10.3	8.7	6.6	8.1	7.3
		MCWBR	9.4	8.9	6.4	6.6	6.5	5.0	4.8	5.9	5.6	4.7	7.9	7.0
Clear-Sky Solar Irradiance	taub	0.255	0.299	0.352	0.376	0.390	0.404	0.420	0.412	0.357	0.328	0.267	0.237	
	taud	1.967	2.013	1.960	2.066	2.278	2.333	2.281	2.268	2.385	2.244	2.029	1.908	
	Ebn at Noon	422	638	728	802	829	823	795	765	750	616	402	261	
	Edh at Noon	43	80	121	132	117	113	116	109	81	64	40	27	
All-Sky Solar Radiation	RadAvg	0.36	1.22	2.83	4.41	4.98	5.31	5.12	3.59	2.18	0.95	0.37	0.17	
	RadStd	0.03	0.13	0.18	0.45	0.41	0.43	0.51	0.37	0.25	0.10	0.08	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page